



Evidence for aerobic ATP synthesis in isolated myelin vesicles

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ABSTRACT

Even though brain represents only 2–3% of the body weight, it consumes 20% of total body oxygen, and 25% of total body glucose. This sounds surprising, in that mitochondrial density in brain is low, while mitochondria are thought to be the sole site of aerobic energy supply. These data would suggest that structures other than mitochondria are involved in aerobic ATP production. Considering that a sustained aerobic metabolism needs a great surface extension and that the oxygen solubility is higher in neutral lipids, we have focused our attention on myelin sheath, the multilayered membrane produced by oligodendrocytes, hypothesizing it to be an ATP production site. Myelin has long been supposed to augment the speed of conduction, however, there is growing evidence that it exerts an as yet unexplained neurotrophic role. In this work, by biochemical assays, Western Blot analysis, confocal laser microscopy, we present evidence that isolated myelin vesicles (IMV) are able to consume O₂ and produce ATP through the operation of a proton gradient across their membranes. Living optic nerve sections were exposed to MitoTracker, a classical mitochondrial dye, by a technique that we have developed and it was found that structures closely resembling nerve axons were stained. By immunohistochemistry we show that ATP synthase and myelin basic protein colocalize on both IMV and optic nerves. The complex of data suggests that myelin sheath may be the site of oxygen absorption and aerobic metabolism for the axons.

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1. Introduction

The mammalian Central Nervous System (CNS) was estimated to consume more than 20% of the body chemical energy. Even though its weight is 2–3% of that of total body (Attwell and Laughlin, 2001), brain consumes 20% of total inspired oxygen (O₂) (Silver and Erecinska, 1998) and 25% of the body's total glucose with an immense metabolic demand (Kann and Kovacs, 2007; Silver and Erecinska, 1998). Neurons critically depend on O₂ supply, being most of neuronal ATP generated by oxidative metabolism (Ames, 2000).

Nevertheless mitochondrial density in brain is lower than in other tissues (Veltri et al., 1990). If we look at axons, mitochondria are few and localized only around in Ranvier's node (Edgar et al., 2008). This sounds paradoxical, considering the exclusivity of mitochondria in aerobic ATP production.

Taken together, these data would suggest that structures other than mitochondria are involved in aerobic ATP production. So we have focused our attention on myelin sheath, the multilayered membrane (Morell and Norton, 1980) produced by oligodendrocytes. Two characteristics of myelin are intriguing: its enormous surface extension, a prerequisite for O₂ absorption, and its high content of neutral lipids (cerebrosides and cholesterol), in which O₂ is known to be absorbed 3–5-fold more readily (Sidell, 1998).

Myelin has long been supposed to augment the speed of conduction (Kursula, 2001), however, there is growing evidence that it exerts an as yet unexplained neurotrophic role. In fact, loss of myelin in demyelinating diseases does not simply cause a lowering of speed of conduction but axonal necrosis (Ferguson et al., 1997). Several authors have investigated the metabolic interaction between neurons and glia (Hertz et al., 2007). Hargittai and Lieberman (1991) observed that myelin contributes approximately 70% of the O₂ consumption in medial giant axon of the crayfish. Recently, lactate produced by astrocytes was hypothesized to be a

Abbreviations: GA, glutaraldehyde; MBP, myelin basic protein; WB, Western Blot; ANT, adenosine nucleotide translocase; AK3, adenylate kinase isoform 3; TIM, mitochondrial import inner membrane translocase; TOM, mitochondrial import outer membrane translocase; PBS, phosphate buffered saline; MW, molecular weight; R.O.D., Relative Optical Density; RH-123, Rhodamine 123; CLSM, Confocal Laser Scanning Microscopy; IMV, isolated myelin vesicles; MT, MitoTracker; MTP, mitochondrial transition pore.

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major neuronal energy substrate (Pellerin and Magistretti, 2004; Schurr, 2006) (astrocyte–neuron–lactate shuttle), but this hypothesis has been challenged (Korf, 2006).

Three proteomic studies are reported in literature, that identified the presence of many subunits of F1-Fo ATP synthase and redox chain, in isolated myelin preparation (Taylor et al., 2004; Vanrobaeys et al., 2005; Werner et al., 2007). Taylor pointed up the connection between energetics and myelin. In our recent proteomic study of the retinal rod outer segment disks (that are devoid of mitochondria), we reported the expression of most of F1-Fo ATP synthase (and redox chain) subunits which was active in ATP production (Panfoli et al., 2008).

2. Materials and methods

2.1. Myelin isolation

Myelin was isolated from five forebrain stems (5 g) of cattle of less than 1 year of age, collected in a slaughterhouse, by the 'floating up' sucrose gradient modification (Haley et al., 1981) of the method by Norton and Poduslo (1973) to reduce contaminants. Protease inhibitor cocktail (Sigma–Aldrich) 50 µg/ml 5-fluorouracil and 20 µg/ml ampicillin were present throughout isolation. Centrifugation was conducted in a Beckman FW-27 rotor (Beckman, Fullerton, CA, USA). *Step 1*: 20 g of sample were homogenized with a Potter-Elvehjem homogenizer in 20 vol. (w/v) of 0.32 M sucrose in 2 mM EGTA. For each tube, 30 ml of homogenate was layered over 28 ml of 0.85 M sucrose in 2 mM EGTA and the tube was centrifuged at 75 000 × g for 30 min. This step was repeat twice. *Step 2*: The layer of crude myelin at the interface of the two sucrose solutions was collected, and homogenized in water to a final volume of 60 ml. Suspension was centrifuged at 75 000 × g for 15 min. *Step 3*: Pellet was dispersed in a total volume of 60 ml of water and centrifuged 12 000 × g for 10 min. The cloudy supernatant was discarded. *Step 4*: The loosely packed pellet was dispersed in water and centrifuged at 12 000 × g for 10 min. The myelin pellet was suspended in 30 ml of 0.32 M sucrose in 2 mM EGTA, and suspension layered onto 0.85 M sucrose in 2 mM EGTA and centrifuged at 75 000 × g for 30 min.

2.2. TEM microscopy

IMV and optic nerves were fixed in 2% paraformaldehyde and 0.2% glutaraldehyde (GA) included in gelatin and freeze-dried in liquid nitrogen. Ultrathin sections (60 nm thick) were obtained with a microtome were put onto classical copper grids. These were labelled with the Ab anti-myelin basic protein (MBP), then anti-rat secondary Ab and protein-A bound to colloidal gold (15 nm). Labelling is fixed with 1% GA.

For colocalization analysis, the IMV and ultrathin optic nerve sections were labelled with the Ab anti-MBP and F1-Fo ATP synthase, then recognized by anti-rat secondary Ab bound to colloidal gold of 10 nm and by anti-rabbit secondary Ab bound to colloidal gold of 15 nm, respectively.

2.3. Electrophoresis, semiquantitative Western Blot (WB) and quantification

Denaturing electrophoresis (SDS–PAGE) was performed using a Laemmli (1970) protocol. Rabbit polyclonal Ab against recombinant human α/β F1-Fo ATP synthase subunits (Panfoli et al., 2008) was diluted 1:5000 in phosphate buffered saline (PBS); rat monoclonal anti-MBP, that identifies an 18–20 kDa band (a gift of Prof. Angelo Schenone, University of Genova); Anti- Na^+/K^+ -ATPase, anti-adenosine nucleotide translocase (ANT) and adenylate kinase isoform 3 (AK3) (Santa Cruz, CA, USA) were diluted 1:400 in PBS.

Anti-mitochondrial import inner membrane translocase (TIM), subunit 8A and mitochondrial import outer membrane translocase, subunit 20 (TOM) (Santa Cruz, CA, USA) were diluted 1:200 in PBS. Secondary Abs were from Sigma–Aldrich. Protein molecular weight (MW) markers were from Fermentas (Fermentas Life Sciences, Hanover, MD, USA). Quantitative densitometry was performed using the ImageJ 1.31v software (<http://rsb.info.nih.gov/ij/>). Each band was converted into a densitometric trace allowing calculations of intensity (Jarsskog and Gilmore, 2000). Enhanced chemiluminescent (ECL) band signals were compared with the signal of whole protein pattern (Ravera et al., 2007a). Results were expressed as Relative Optical Density (R.O.D.). Quantification data come from five different bovine forebrains.

2.4. Biochemical assays

Na^+/K^+ -ATPase was determined as the difference between the ATPase activities in the presence and in the absence of 0.1 mM ouabain (Balestrino et al., 1998). AK3 assay was performed according to Tomasselli et al. (1979).

2.5. Oxygraphic measurements

O_2 consumption was assayed in forebrain homogenate, crude myelin and isolated myelin, by a thermostatically controlled oxygraph apparatus equipped with an amperometric electrode (Unisense–Microrespiration, Unisense A/S, Denmark) and a rapid mixing device. In a typical experiment, sample was incubated, at 25 °C, in the following respiration solution: 120 mM KCl, 2 mM MgCl_2 , 1 mM KH_2PO_4 , 50 mM Tris–HCl, pH 7.4 and 25 µg/ml ampicillin (final volume 1.7 ml) (Sgobbo et al., 2007). After 5 min, to observe the uncoupled respiration rate, 30 µM Nigericin was added before the addition of respiring substrates and inhibitors of O_2 consumption: 0.7 mM NADH, 20 mM succinate, 10 mM ascorbate, 40 µM rotenone, 50 µM antimycin A and 0.5 mM potassium cyanide. To observe the ADP-stimulated respiration rates 0.045 mM ADP was added after NADH and succinate addition. Mitochondria-enriched fraction was used as a positive control, in this case NADH was substituted by 5 mM pyruvate and 2.5 mM malate. The respiratory rates were expressed in µM O_2 /min/mg.

2.6. Assay of redox complexes

All the four mitochondrial complexes were assayed according to Sottocasa et al. (1967), Janssen et al. (2007) and Baracca et al. (2003).

2.7. Spectrofluorimetric measurements of proton potential

Proton potential ($\Delta\mu_{\text{H}^+}$) was assayed using Rhodamine 123 (RH-123) dissolved in ethanol (below 0.4%). Fluorescence intensity (excit.: 503; emiss. 527 nm) was monitored at 25 °C with a PerkinElmer LS 50B fluorimeter (Emaus et al., 1986) under continuous stirring. Sample (20 µg) was incubated for 5 min in: 0.32 M sucrose, 0.01 mM HEPES, 2 mM MgCl_2 , 4 mM NaH_2PO_4 , 5 mM KCl 10 mM glucose and 2.5 U/ml hexokinase (ADP regenerating system). Then 50 nM RH-123 was added to the mixture and $\Delta\mu_{\text{H}^+}$ changes monitored. Additions were: 0.1 mM ADP, 0.01 mM rotenone (for four time), 20 mM succinate (for three time), 0.1 µM oligomycin (for four time) and 0.04 mM antimycin A (Baracca et al., 2003).

2.8. RH-123 fluorescence quenching rate

F1-Fo ATP synthase concentration was estimated by RH-123 fluorescence quenching rate using different concentration of oligomycin. Reaction mixture contained: 20 µg of protein, 0.32 M sucrose, 0.01 mM HEPES, 2 mM MgCl_2 , 4 mM NaH_2PO_4 , 5 mM KCl,

10 mM glucose and 2.5 U/ml hexokinase (for ADP regenerating system), 0.1 mM ADP, 0.01 mM rotenone, 20 mM succinate. Then, different concentrations of oligomycin were added. Fluorescence quenching was monitored with an PerkinElmer LS 50B fluorimeter (503 nm excit., 527 nm emiss.) (Baracca et al., 2003).

2.9. ATP synthase assay

ATP synthesis rate was measured by the luciferin/luciferase chemiluminescent method (Roche Applied Science), according to Sgarbi et al. (2006).

2.10. Confocal Laser Scanning Microscopy (CLSM)

CLSM imaging was performed as previously described (Ravera et al., 2007b; Bianchini et al., 2008).

2.11. Immunohistochemistry on optical nerve

The bovine optic nerve was fixed in 4% paraformaldehyde in phosphate-buffered saline (PBS, pH 7.4) for 12 h. After careful rinses in PBS, the specimen was dehydrated and embedded in Paraplast. Longitudinal Paraplast sections (5 μ m thick) were cut and rehydrated in a decreasing ethanol series and washed in PBS (0.1 M, pH 7.4).

Double-immunohistochemical staining techniques were carried out according to the indirect immunofluorescence method. Briefly, dewaxed serial sections were incubated overnight in a moist chamber at 4 °C with rabbit polyclonal Ab against β subunits of F1-Fo ATP synthase diluted with PBS 0.1% bovine serum albumin (BSA), then, after several washing in PBS, the sections were incubated with Alexa-488-conjugated anti-rabbit antiserum (1:800; Molecular Probes, USA), for 1 h. After, the same sections were incubated overnight, with a rat polyclonal Ab against MBP (1:200), and a secondary Cy3-conjugated anti-rat Ab (1:800; Sigma-Aldrich). The sections were then stained with DAPI, thoroughly washed in PBS and mounted in a glycerol/PBS (1:1) solution and examined under a Olympus BX60 epifluorescence microscope.

2.12. Other procedures

Mitochondria-enriched fraction from bovine liver was obtained according to Ravera et al. (2007a).

Protein concentration was determined by Bradford method, using BSA as a standard protein (Bradford, 1976).

3. Results

3.1. Myelin characterization

Fig. 1 shows the characterization of the forebrain stem-derived samples (forebrain homogenate, crude myelin fraction, isolated myelin fraction), as well as of mitochondria-enriched fractions, used as a control. Panel A shows the morphology of IMV, as observed by TEM. The sample appears to be composed by numerous vesicles, some maintaining the multilamellar structure typical of myelin sheath, some other being uni-lamellar. Immuno-gold labelling technique (Panel B), shows that myelin vesicles are decorated by an Ab against MBP. Panel C shows the protein pattern as stained with Colloidal Blue Coomassie. Semiquantitative WB analyses were conducted with antibodies against MBP, and Na^+/K^+ -ATPase, a marker of neurilemma (Panels D and E). Panel D shows the chemiluminescent WB signal of MBP, only detectable in forebrain-derived samples, whose intensity increased in parallel to the isolation grade of myelin fractions. Panel E shows that Na^+/K^+ -ATPase was present

in mitochondria-enriched fraction (probably due to a contamination from plasma membrane) and in forebrain homogenate, less in crude myelin and absent in isolated myelin. Data are confirmed by the densitometric analysis of chemiluminescent signal, reported in Panel F. Na^+/K^+ -ATPase activity was monitored in forebrain-derived samples by a spectrophotometrical assay (Table 1). Loss of Na^+/K^+ -ATPase amounted to 82% in crude myelin and 91.4% in IMV, with respect to enzyme activity in forebrain homogenate.

3.2. Respiration rate in isolated myelin vesicles

To verify the physiological implications of our data, we conducted an analysis of respiratory fluxes in IMV fraction. Fig. 2, Panel A, reports a typical amperographic tracing of basal and Nigericin-uncoupled respiration rate. Following Nigericin incubation, the maximal respiratory fluxes by NADH (Complex I + III + IV), succinate (Complex II + III + IV) and ascorbate (Complex IV) were measured as rates sensitive to inhibition by rotenone, antimycin A and cyanide, respectively. Fig. 2, Panel B reports a typical amperometric recording of ADP-stimulated respiration rates. After recording the endogenous respiration rate, NADH and succinate were added as respiratory substrates followed by ADP addition. This induced a transient stimulation of the O_2 consumption rate. Respiration rates decreased by addition of rotenone and antimycin A, respectively. These conditions allowed to determine the oxidative phosphorylation efficiency (P/O ratio), which is measured as coupling of O_2 consumption to ATP production. The P/O ratio was calculated as the ratio between the amount (nanomoles) of added ADP and the O_2 (nanomoles) consumed during the ADP-induced state 3 respiration. P/O ratio with NADH was 2.44 ± 0.24 ; and with succinate 1.68 ± 0.17 . The little difference among O_2 consumption in IMV with respect to forebrain homogenate, would indicate that the former are the major O_2 consumer, even though in latter sample, mitochondria are present.

Table 2 reports a comparison of oxygen consumption, in uncoupled conditions, among mitochondria and the forebrain-derived samples. For the latter, NADH (Complex I + III + IV) and succinate (Complex II + III + IV) were used as substrates, while for mitochondria NADH was substituted with pyruvate/malate (2.5 and 5 mM, respectively).

3.3. Redox complex assays and semiquantitative WB analysis

Fig. 3 shows the activities of the redox complexes in mitochondrial enriched fraction (using a positive control) and in all the forebrain-derived samples, in absence or presence of specific inhibitors (rotenone for NADH-Ubiquinone oxidoreductase, 73% inhibition, Panel A; antimycin A for Cytochrome c reductase, 75% inhibition, Panel C; cyanide for Cytochrome c oxidase, 99% inhibition, Panel D). Succinic dehydrogenase was assayed in two different ways (Panel B). Activity of each complex in myelin fraction was comparable to that of the forebrain homogenate. In Panels E–H, semiquantitative WB analysis is reported, Abs were against NADH-Ubiquinone oxidoreductase (ND4L), Cytochrome c and Cytochrome c oxidase.

3.4. Assay of proton gradient on MV

The activity of the redox chain generates a proton gradient ($\Delta\mu_{\text{H}^+}$) across the membrane, producing a pH gradient and a membrane potential, used by F1-Fo ATP synthase to produce ATP. The presence of a $\Delta\mu_{\text{H}^+}$ on myelin vesicle membranes was assayed by monitoring fluorescence quenching of RH-123. Usually this dye is used to monitor the membrane potential of mitochondria, as energization induces quenching of RH-123 fluorescence and the rate of fluorescence decay is proportional to the mitochon-

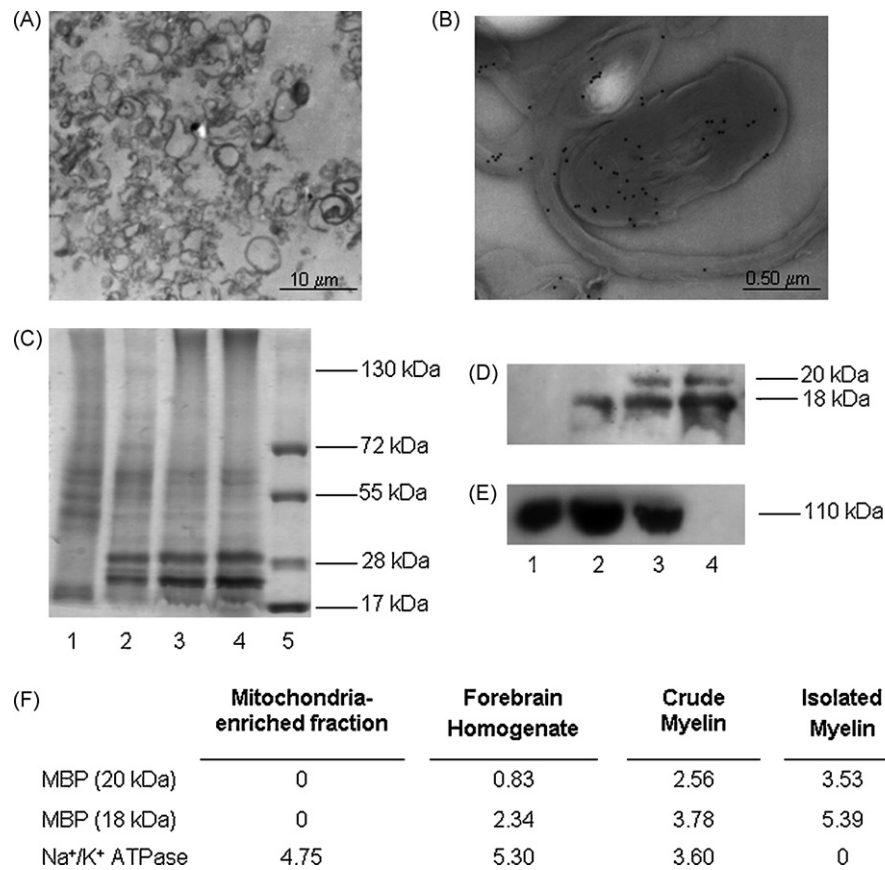


Fig. 1. Characterization of forebrain-derived samples. Panel A is a TEM image of the IMV suspension. The medium length of the membrane profiles ranged around 1.4 μm. Higher magnifications (Panel B) show the immuno-gold analysis of a single multilamellar myelin vesicles, marked with an antibody against MBP. Signal is uniformly diffused on the vesicles. Both panels are representative of 10 different fields. Panel C shows the protein pattern of samples (lane 1, mitochondria-enriched fraction; lane 2, forebrain homogenate; lane 3, crude myelin fraction; lane 4, isolated myelin fraction; lane 5, molecular weight markers), stained with Colloidal Blue Coomassie. Panels D and E show the semiquantitative Western Blot of MBP and Na⁺/K⁺-ATPase, respectively. The ratios of these signals were analyzed by densitometry, showed in Panel F. The values of densitometric analysis are expressed as Relative Optical density (R.O.D.). Each panel is representative of at least five experiments.

Table 1
Na⁺/K⁺-ATPase activity.

	Mitochondria-enriched fraction	Forebrain homogenate	Crude myelin	Isolated myelin
Na ⁺ /K ⁺ -ATPase (mU/mg)	12.3 ± 0.68	15.7 ± 0.8	2.8 ± 0.14	1.35 ± 0.06

The table reports the assay of Na⁺/K⁺-ATPase, a typical plasma membrane protein, in forebrain-derived samples and mitochondria-enriched fraction. Activity was expressed in mU/mg of protein (nmol ATP hydrolyzed/min/mg). Each assay is representative of at least eight experiments and data are represented as mean ± SD.

drial membrane potential (Baracca et al., 2003). Fig. 4, Panel A shows the effect of consecutive additions on the dye fluorescence: 0.02 mg/ml of myelin to the medium containing 50 nM RH-123 and ADP regenerating system induced a little enhancement of the RH-123 fluorescence. An enhancement of fluorescence was observed by adding ADP, indicating that the Δμ_{H+} decreased due to ADP phosphorylation. Rotenone caused a further fluorescence increase, likely due to the membrane potential dissipation. Potential could be

recovered by addition of succinate, which induced the quenching of the dye. Finally, oligomycin induced an increase in the membrane potential whereas addition of antimycin A allowed a recovery of fluorescence caused by membrane potential decrease, likely as a consequence of proton gradient dissipation.

Fig. 4, Panel B shows the quantification of F1-Fo ATP synthase by the inhibitory effect of oligomycin on the respiration-induced initial fluorescence quenching rate of RH-123. At low inhibitor

Table 2
Oxygen consumption in derived forebrain samples and mitochondria.

Samples	Oxygen consumption (μM O ₂ /min/mg)		
	Complex I + III + IV	Complex II + III + IV	Complex IV
Forebrain homogenate	42.37 ± 2.12	64.95 ± 3.24	38.25 ± 1.91
Crude myelin	38.54 ± 1.93	62.05 ± 3.11	35.4 ± 1.77
Isolated myelin	39.06 ± 1.97	61.38 ± 3.14	35.12 ± 1.76
Mitochondria	60.63 ± 3.03 ^a	91.58 ± 4.58	52.68 ± 2.64

The table shows a comparison of respiration rates of forebrain homogenate, crude myelin, isolated myelin and mitochondria, on the basis of three pathways: Complex I + III + IV, Complex II + III + IV, only Complex IV. O₂ consumption is expressed as μM O₂/min/mg of total protein.

^a In mitochondria, to stimulate the Complex I + III + IV pathway, 5 mM pyruvate and 2.5 mM malate substituted for NADH.

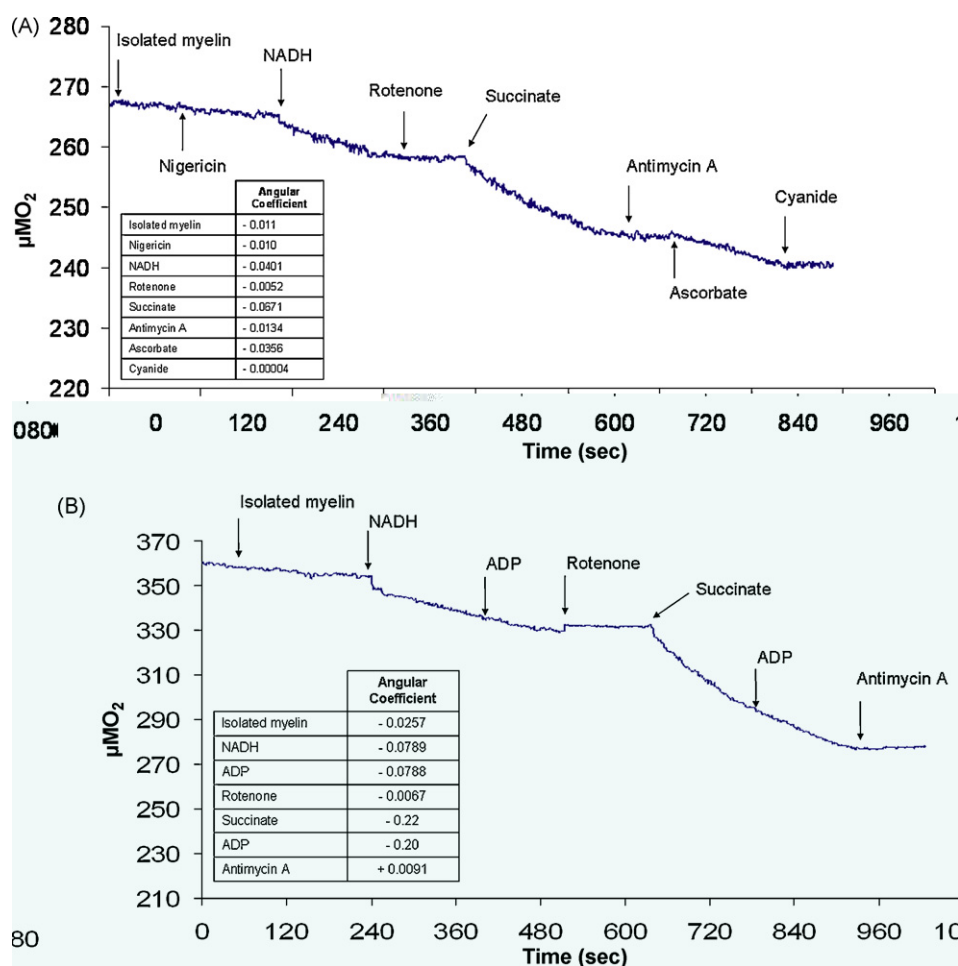


Fig. 2. Measurements of respiration rates in IMV. Panel A reports the amperometric recording of basal and Nigericin-uncoupled respiration rates by the addition of several respiring substrates. The inset shows the changes of the angular coefficient values, during the addition of several substrates. Panel B reports the amperometric recording of ADP-stimulated respiration rates. Inset shows the changes of the angular coefficient values, during the addition of several substrates. Each panel is representative of at least five experiments.

concentrations, fluorescence decreased as oligomycin concentration increased, until a plateau was reached at about 50 nM. The quenching rate showed a hyperbolic dependence on oligomycin concentration. By plotting data in the double-reciprocal form (Fig. 4, Panel B, inset), the intercept of the straight line with the abscissa was at $-221.87 \mu\text{M}^{-1}$ corresponding to a concentration of 4.5 nM oligomycin necessary for 50% fluorescence quenching rate increase. The F1-Fo ATP synthase/oligomycin stoichiometry is 1 to 1, and binding is irreversible. The expected concentration of F1-Fo ATP synthase in the fluorimeter cuvette is 9 nM, corresponding to 0.9 nmol F1-Fo ATP/mg of IMV protein.

3.5. ATP synthesis in MV

The rate of ATP synthesis was measured in all the forebrain-derived samples prior energized with succinate. In order to assess whether ATP synthesis was carried out by F1-Fo ATP synthase we included in the medium adenylate kinase inhibitors, and exogenous Mg^{2+} , in order to supply saturating Mg-ADP to the ATP synthase complex.

Fig. 5, Panel A reports the ATP synthase activity in IMV. This activity was inhibited by oligomycin (88%), the F1-Fo ATPase/ H^+ -pump inhibitor, DCCD (99%), used as inhibitor of passive H^+ conduction and antimycin A (70%), a specific inhibitor of Complex III. In Fig. 5, Panel B, a comparison among the forebrain-derived samples and the mitochondrial enriched fraction is reported. Isolated myelin

produces more ATP than the other samples. The mitochondria-enriched fraction was only able to produce ATP in presence of Cyclosporin A or 2-Aminoethoxydiphenyl borate. A semiquantitative WB analysis, using an Ab against α/β subunits of F1-Fo ATP synthase (Panel C) of samples as in Fig. 1 shows (Panel D) that the protein is more enriched in isolated myelin.

3.6. Evaluation of the results in terms of the entity of mitochondrial contamination

The procedure to obtain myelin fraction has been carefully designed. Most of mitochondrial fraction was eliminated during the first step. The forebrain-derived sample was prepared in absence of Cyclosporin A (Crompton, 1999) and 2-Aminoethoxydiphenyl borate (Chinopoulos et al., 2003), in order to support the formation of transition pore (TP) in mitochondria. A semiquantitative WB with an Ab against ANT and TIM (two typical inner membrane proteins) and TOM (a typical outer membrane protein) was carried out. The WB signals of the cited proteins were only detectable in mitochondrial enriched fraction and forebrain homogenates (Fig. 6, Panels A–C). Data are confirmed by densitometric analysis (Panel D). AK3, a typical mitochondrial matrix enzyme was also assayed. Table 3 shows that AK3 activity was present in mitochondria-enriched fractions and in the pellet obtained after the first step of myelin isolation, decreased in forebrain homogenate and was absent in isolated myelin fractions.

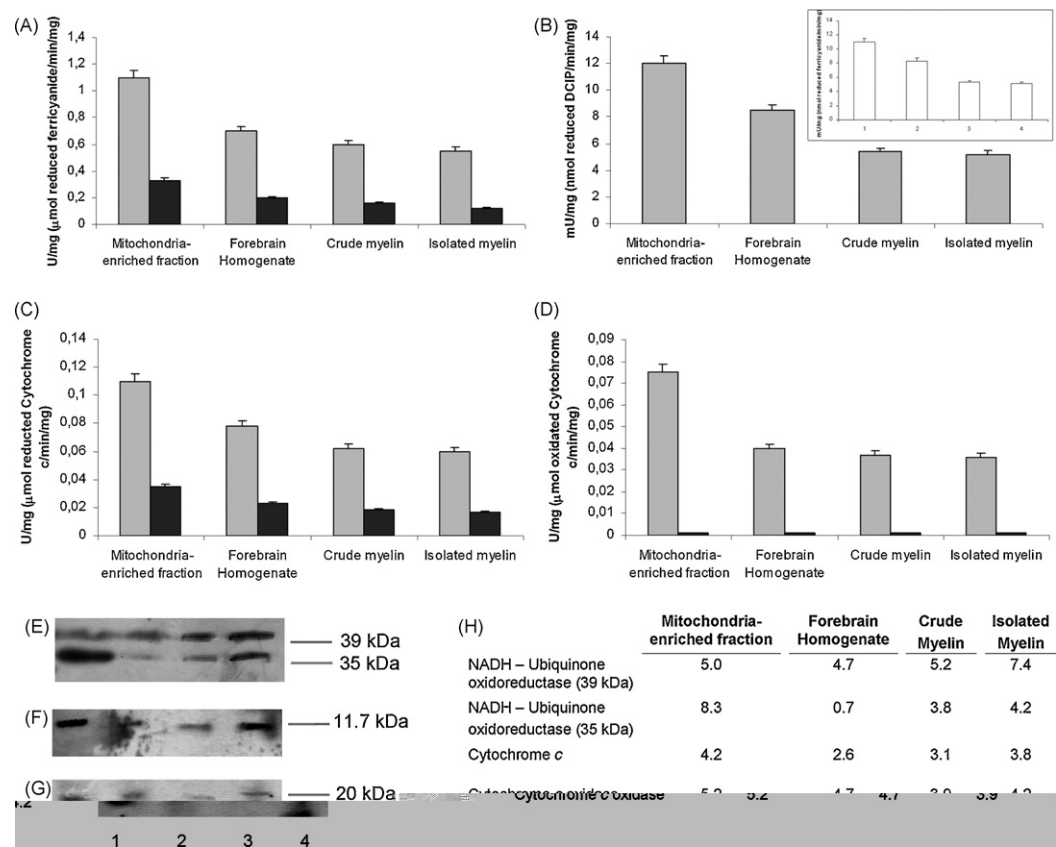


Fig. 3. Respiration Complexes Assays and semiquantitative WB analysis. Panels A–D report a comparison of the activity of the redox Complexes I–IV (respectively) in forebrain-derived samples and in mitochondria-enriched fractions. The black columns show the activity in the presence of the specific inhibitors. Each panel is representative of at least ten experiments. In Panels E–H, semiquantitative WB analysis is shown. Samples are: Lane 1, mitochondria-enriched fraction; lane 2, forebrain homogenate; lane 3, crude myelin fraction; lane 4, isolated myelin fraction. Panels E–G report the NADH-Ubiquinone oxidoreductase (ND4L), Cytochrome c and Cytochrome c oxidase chemiluminiscent signals, respectively. The ratios of these signals were analyzed by densitometry, showed in Panel H. Each panel is representative of at least five experiments.

3.7. Confocal imaging

By a new technique that we have developed (Bianchini et al., 2008), IMV and optical nerves were incubated with MitoTracker (MT) Deep Red 633, a fluorescent dye that stains mitochondria in living cells and its accumulation is dependent upon membrane potential. After incubation, the samples were analyzed by CLSM. Normally, this probe stains only mitochondria, but in our case also the myelin vesicles became fluorescent (Fig. 7, Panel A), indicating that myelin sheath is an actively respiring structure. Accordingly, also optical nerve sections were stained with this probe, and a 3D analysis of this image showed a tubular structures, which were uniformly stained and had a diameter of about 1 μm (Fig. 7, Panels B and C).

3.8. Colocalization of MBP and F1-Fo ATP synthase on IMV and optic nerve

Fig. 8 shows the colocalization of MBP and F1-Fo ATP synthase on IMV (Panel A) and optical nerve sections (Panels B and C), as detected by TEM. In both samples the signal of Ab against MBP (col-

loidal gold of 10 nm) coexist with the Ab against F1-Fo ATP synthase (colloidal gold of 15 nm).

Fig. 9 shows a longitudinal section of optic nerve with Ab against F1-Fo ATP synthase (Panel A) and MBP (Panel B). Proteins are stained by indirect fluorescence being secondary Ab for F1-Fo ATP synthase conjugated to Alexa-488 (green) and secondary Ab for MBP conjugated to Cy3 (red). To observe the nuclei, the section was stained with DAPI (Panel C). Panel D shows the image of the merge fluorescence of Panels A–C.

4. Discussion

The present metabolic biochemical study and the *ex vivo* imaging studies suggest that IMV consume O₂ by an unexpected operativity of the redox chain (Complexes I–IV + F1-Fo ATP synthase) for ATP synthesis in IMV, i.e. outside the mitochondria. O₂ consumption and ATP synthesis by myelin (Fig. 2) seem specific, being sensitive to the common inhibitors of mitochondrial redox Complexes I–V.

Interestingly, the basal and Nigericin-uncoupled respiration rate of IMV are similar to those observed in mitochondria (Table 2),

Table 3
Adenylate kinase, isoform 3 activity.

	Mitochondria-enriched fraction	Forebrain homogenate	Pellet after step 1 of myelin isolation	Crude myelin	Isolated myelin
AK3 activity (U/mg)	1.4 ± 0.07	0.017 ± 0.008	0.013 ± 0.006	0.005 ± 0.002	0

The table reports the assay of AK3, a typical mitochondria-matrix protein, in forebrain-derived samples and mitochondria-enriched fractions. Activity was expressed in U/mg of total protein (μmol ADP produced/min/mg). Each assay is representative of at least eight experiments and data are represented as mean ± SD.

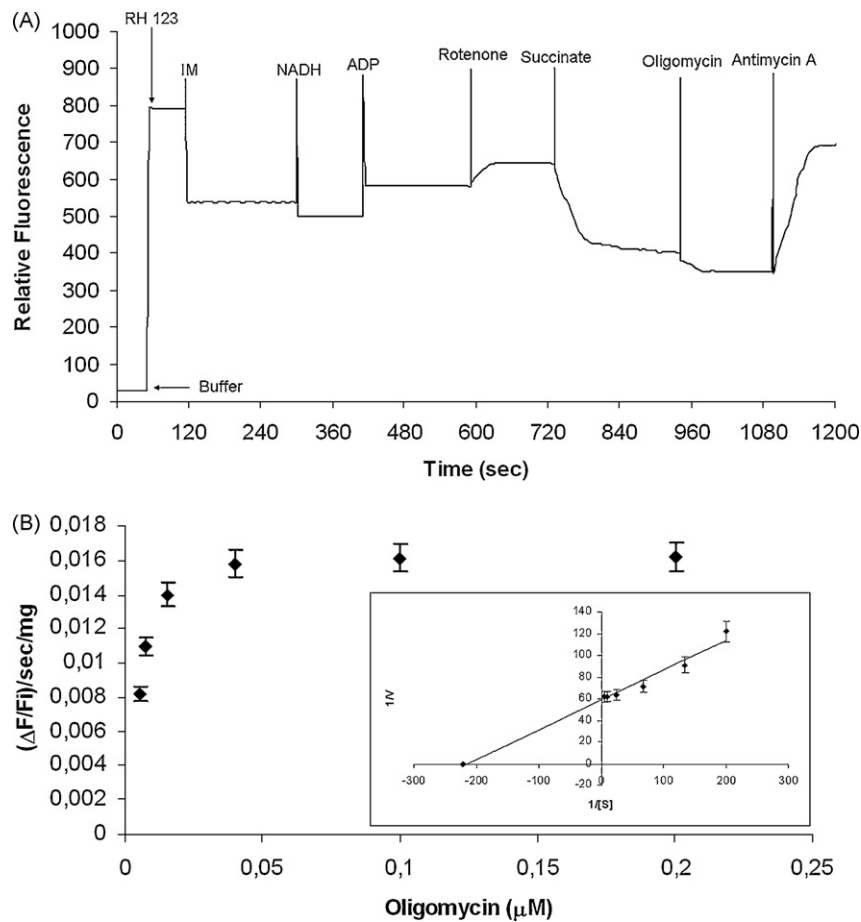


Fig. 4. Determination of proton gradient presence on IMV surface. Panel A reports the time course of RH-123 fluorescence upon addition of several substrates and inhibitors of oxidative phosphorylation in IMV. Fluorescence was measured on a spectrofluorimeter, by exciting at 503 nm and collecting the emitted fluorescence at 527 nm. Panel B shows the titration of RH-123 fluorescence quenching rate with oligomycin. Initial rate was measured after 30 $\mu\text{g}/\text{ml}$ IMV suspended in the respiratory buffer. Oligomycin was added at the indicating concentration. The estimated rate in the absence of oligomycin was subtracted from each value at the different inhibitor concentrations. Inset represents the double-reciprocal plot, from which the F₁-F₀ ATP synthase concentration was estimated. Each panel represents at least five experiments.

but are not sensitive to the potential loss that mitochondria suffer. In fact, the forebrain-derived samples were prepared in the absence of Cyclosporin A and 2-Aminoethoxydiphenyl borate, which inhibit the contact between ANT and cyclophilin (Crompton et al., 1988) causing the opening of the mitochondrial permeabil-

ity transition pore (MTP) (Beutner et al., 1996), release of matrix components and eventual collapse. Mitochondria functioning need a coupled mitochondrial inner membrane (Mitchell, 1961; Boyer, 1997). Studies on sciatic nerves demonstrated that myelin sheath absorbs O₂ (Hargittai and Lieberman, 1991). Activities of the four

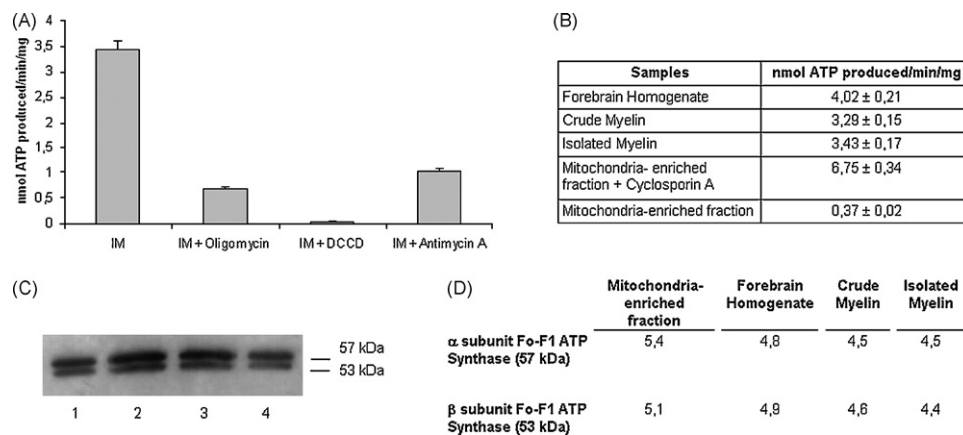


Fig. 5. ATP synthesis in IMV and semiquantitative WB analysis. Panel A shows the ATP synthesis assay on IMV in the absence/present of the common inhibitors. Inhibition by oligomycin (0.002 mM), DCCD (1 mM) and antimycin A (0.02 mM) was around 80%, 99% and 70% respectively. Panel B compares ATP synthesis in forebrain-derived samples and in mitochondria-enriched fraction. Mitochondria-enriched fraction was assayed in the presence of Cyclosporin A. In both panels, activity is expressed as nmol ATP produced/min/mg of total protein. Panels C reports the semiquantitative WB analysis of α/β subunits of F₁-F₀ ATP synthase in forebrain-derived samples and mitochondria-enriched fraction. Samples are: Lane 1, mitochondria-enriched fraction; lane 2, forebrain homogenate; lane 3, crude myelin fraction; lane 4, isolated myelin fraction. Panel D shows the densitometric analysis of the signal. Each panel is representative of at least ten experiments.

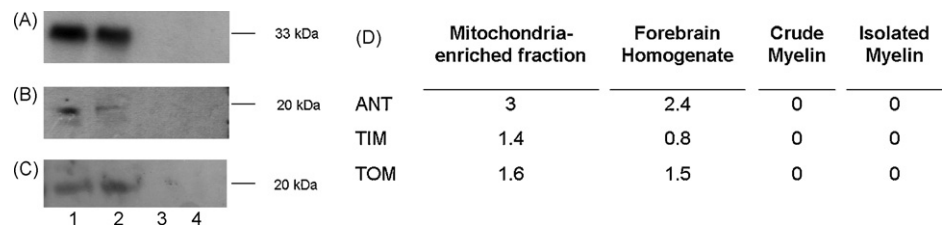


Fig. 6. Evaluation of the results in terms of the entity of mitochondrial contamination. This figure shows a semiquantitative WB with an antibody (Ab) against ANT and TIM (two typical mitochondrial inner membrane proteins) and TOM (a typical mitochondrial outer membrane protein) on forebrain-derived samples and mitochondria-enriched fraction (Panels A–C, respectively). Samples are: Lane 1, mitochondria-enriched fraction; lane 2, forebrain homogenate; lane 3, crude myelin fraction; lane 4, isolated myelin fraction. Panel D reports the densitometric analysis of these signals (Panel D). Each panel is representative of at least five experiments.

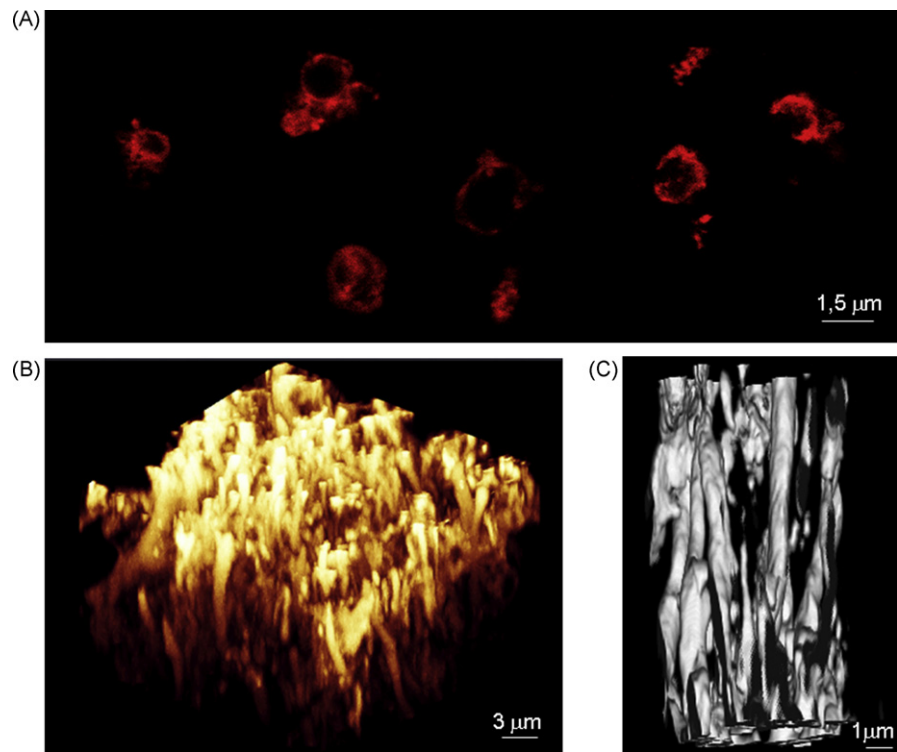


Fig. 7. Confocal imaging of and optic nerve section stained with MitoTracker. Panel A reports the CLSM image of IMV stained with MitoTracker (MT) Deep Red 633. Scale bar is 1.5 μm . Panels B and C show a CLSM images of MT fluorescence in a optic nerve section. Scale bars are 3 and 1.5 μm , respectively. In particular in Panel B, a 3D image of an optic nerve section (20 μm thickness) is reported. Panel C shows a 3D reconstruction of a portion of the image stack visualized in Panel B. In both images, tubular structures were visible; these structures had a medium diameter of about 1 μm , which corresponds to the dimension of the myelinated axon in optic nerve. Each panel is representative of at least four experiments.

redox complexes in IMV were comparable to that of mitochondria (Fig. 3) and sensitive to the specific inhibitors. Moreover, myelin sheath contains Cardiolipin (Heape et al., 1986), a lipid almost exclusive of the mitochondrial inner membrane (Wang et al., 2007) necessary for Cytochrome c oxidase function (Zhang et al., 2005). It was also reported that CL acts as a proton (H^+)-

trap (Haines and Dencher, 2002). The expression ($\sim 30\%$ of del total) of many of the redox complexes that usually have a mitochondrial location has been previously reported in myelin (Taylor et al., 2004; Vanrobaeys et al., 2005; Werner et al., 2007), but the reason for the presence of these proteins has remained elusive.

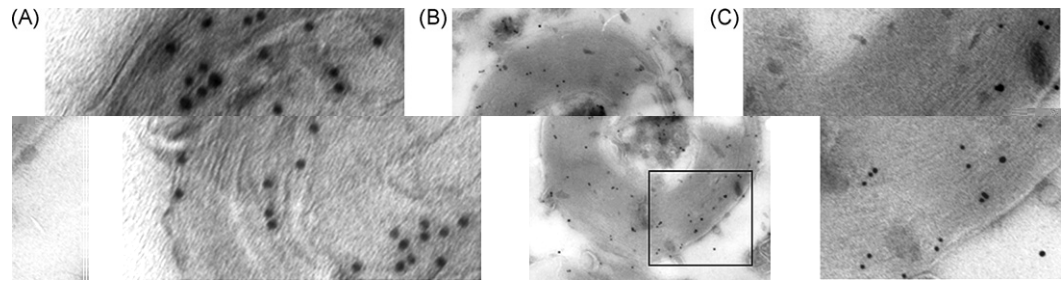


Fig. 8. Colocalization of MBP and F1-Fo ATP synthase by TEM. The colocalization of MBP and F1-Fo ATP synthase on IMV (Panel A) and optical nerve sections (Panel B). A zoomed view of the section in Panel B is shown in a black square (Panel C). The MBP signal is represented by the colloidal gold of 10 nm diameter, and F1-Fo ATP synthase is represented by the colloidal gold of 15 nm diameter. The two signals coexist both on IMV and on optical nerve sections.

Fig. 9. Colocalization of MBP and F1-Fo ATP synthase on longitudinal optic nerve sections. A longitudinal section of optic nerve immunohistochemically stained with both an Ab against F1-Fo ATP synthase (Panel A) and MBP (Panel B). Proteins are stained by indirect fluorescence being secondary Ab conjugated to Alexa-488 (green) (for F1-Fo ATP synthase) and secondary Ab conjugated to Cy3 (red) (for MBP). Panel C shows the DAPI stain. The image of the merged fluorescence of Panels A–C is in Panel D. The yellow signal indicates the colocalization between the signal of MBP and of F1-Fo ATP synthase.

The operation of redox complexes generates a transmembrane electrochemical $\Delta\mu_{H^+}$ whose presence in IMV is suggested by fluorimetric analysis of RH-123 redistribution across the membranes of IMV (Fig. 4A). Interestingly, myelin is unusually impermeant to H^+ (Diaz and Monreal, 1994), a prerequisite for the formation of a $\Delta\mu_{H^+}$ across the membrane. The presence of $\Delta\mu_{H^+}$ seems confirmed by the CLSM imaging analysis (Fig. 7), where, quite unexpectedly, both myelin vesicles and optic nerve sections were stained by MT, a fluorescent mitochondrial probe that stains mitochondria in living cells and its accumulation is dependent upon membrane potential (Poot et al., 1996). Morphology of IMV in CLSM was clear enough to allow unambiguous identification. When the same analysis was conducted on optic nerve sections, impressive results were obtained by a 3D analysis (Fig. 7). Tubular structures of about 1 μm diameter (the typical diameter of optic nerve axons) extending for the whole section thickness (about 20 μm) appear stained by MT. Fluorescence was uniformly distributed on the axons, as can be seen by the [Supplementary Movie](#). Data would suggest that living optic nerve myelin is actively respiring. If fluorescence was due to the axonal mitochondria, staining would have been restricted to Ranvier's nodes, where they are mostly located. Also, a careful observation of Fig. 9 suggests that the fluorescent structures do not resemble mitochondria, but rather the protein skeleton of myelin sheath. In fact, the processing of the sample implies the loss of most of lipid fraction.

The consistent ATP synthesis by IMV, sensitive to common F1-Fo ATP synthase inhibitors (Fig. 5), would suggest that the ATP synthase operates through the generation of a $\Delta\mu_{H^+}$, like the in mitochondria. The functional localization of F1-Fo ATP synthase in IMV would seem confirmed by semiquantitative WB analysis, as signal increased with the isolation step; and by the observation that the concentration of F1-Fo ATP synthase in IMV (0.9 nmol/mg of myelin protein), as calculated by the quenching of RH-123 fluorescence (Fig. 4B), is similar to that reported for mitochondria, by an

identical technique (Baracca et al., 2003). A quantitative consideration of the entity of ATP synthesis would also rule out the hypothesis of a mitochondrial contamination. Moreover, Figs. 8 and 9 show that F1-Fo ATP synthase and MBP colocalize onto both IMV and optic nerve sections suggesting that the enzyme resides in myelin. An ectopic location of F1-Fo ATP synthase in many mammalian cell membranes has been recently reviewed (Chi and Pizzo, 2006), even though most authors ascribe to it functions other than ATP synthesis. An extracellular ATP synthesis was also reported by the Umbilical Vein Endothelial Cells (HUVECs) that however was not entirely inhibited by oligomycin (Arakaki et al., 2003), and by hepatocyte membranes (Mangiullo et al., 2008). In a recent proteomic study, we reported a functional expression of F1-Fo ATP synthase on the retinal rod outer segment disks (a subcellular fraction devoid of mitochondria) (that is part of CNS) (Panfoli et al., 2008).

Our data seem to demonstrate that ATP can be aerobically produced also by structures other than mitochondria, which we may define “mitochondrial inner membrane-like membranes”. In fact, we hypothesize that the respiration chain present in myelin sheath is the same as expressed in mitochondria. In fact, the antibodies against the redox chains I to V recognized the proteins in both myelin and mitochondrial membranes, indicating a sequence homology. Interestingly, Complex I in IMV is recognized by an Ab against the N4DL subunit, which is coded by mitochondrial DNA. In fact, no exportation sequence has ever been described for mitochondrial DNA through TIM/TOM. On the other hand, the redox Complexes I, III and IV are thought to form a super-complex *in vivo*. So, the assembly of the respiratory chain would take place in mitochondria, with the participation of the same subunits encoded by the nuclear genome, undergoing a biogenesis as a mitochondrial inner membrane. Then, mitochondria may fuse with myelin sheath during its formation, allowing the transfer of the “ready-to-use” complexes to sheath forming “mitochondrial inner membrane-like membranes”. Further studies are necessary to understand how

this happens. An intriguing clue comes from the knowledge that a form of the inherited motor and sensory neuropathy Charcot-Marie-Tooth includes mutations (GDAP1, YARS, and the pleckstrin homology domain of dynamin 2) affecting the mitochondrial fission (reviewed in: Berger et al., 2006). In fact, mitochondria are dynamic organelles which undergo fusion, division, and directed movement in many cell types (reviewed in: Hales, 2004). Mitochondria adopt a variety of different shapes in eukaryotic cells, ranging from multiple, small compartments to elaborate tubular networks (Shaw and Nunnari, 2002). The knowledge of mitochondrial matrix proteins to be exportable is not a new finding, as several mechanisms have been described (Soltys and Gupta, 2000). Examples of mitochondrial matrix proteins found at “extramitochondrial” locations exist (Soltys and Gupta, 1999, 2000). Particularly interesting seem the possibility of a fusion among mitochondria and other membranes (Soltys and Gupta, 1999) including the nucleus, endoplasmic reticulum (ER) and chloroplasts, and with the plasma membrane (Crotty and Ledbetter, 1973).

The possibility that our surprising results are due to never reported physiological fusion of mitochondrial vesicles after their transport to distal parts after their fission from the main mitochondrial network reticulum would be underlined by the colocalization of MBP with F1-Fo ATP synthase (Figs. 8 and 9).

The intriguing hypothesis arises that myelin be a respiring wrap also *in vivo* to supply ATP for the axon. In view of this, the function of the peculiar multilamellar structure of myelin, would become clear: a wide surface development is in fact typical of mitochondrial cristae, for the withdrawal of O₂ from cytosol into the membranes where Cytochrome c oxidase is embedded. Moreover, the cited hypothesis would explain the conspicuous presence in myelin of neutral lipids that otherwise would represent an enormous sink for O₂. Interestingly, the bigger the axon the more numerous the spires (Baumann and Pham-Dinh, 2001). Obviously, such hypothesized role of myelin would not exclude the contribution of axonal mitochondrial, that however are mostly localized in the Ranvier nodes. In fact, a tissue-specific organization of the mitochondrial network as long interconnected cables could be a mean to increase the efficiency of ATP production (Hales, 2004). Likely also astrocytes contribute with anaerobic glycolysis forming lactate (Pellerin and Magistretti, 2004).

Studies on the CNS of the neonatal mammals, compared with the adult, showed that only the first is resistant to anoxia (Duffy et al., 1975). Supposing that the O₂ absorbed by myelin produces ATP through respiration, this would explain why only myelinated axons are sensitive to anoxia: these would be energetically assisted, and absorb more O₂ than the physiologically unmyelinated ones which would not respire through myelin. In unmyelinated axons such as in autonomous nervous system, where conduction is 10-fold lower than in myelinated axons, ATP need is lower.

Overall, data suggest that a mitochondrial contamination would play a negligible role. In fact, the quality of the myelin preparations seems good with minimal contamination by other subcellular fractions (Figs. 1 and 6). On the other hand, unless most of our IMV preparations were composed by intact mitochondria, not visible by TEM analysis (Fig. 1), a contamination may not account for the consistent activity of the five respiratory complexes that we have observed. Also, the absence of ANT, TIM and TOM, and of AK3 activity (Fig. 6A–D and Table 3) suggest that whole mitochondria are not present in the IMV. Particularly, absence of AK3 also indicates that the presence of mitochondrial matrix is negligible, likely due to the procedure employed for isolation of IMV. It is tempting to presume that the absence of ANT may physiologically be supplied by the myelin gap-junctions formed by Connexins, that were shown to be highly permeable to ATP (Goldberg et al., 2002), whose role is not thoroughly understood. It was also shown that the radial pathway of ATP is approximately one million times faster than the circum-

ferential pathway (Balice-Gordon et al., 1998). Mitochondrial redox Complexes I–V, were found enriched in parallel with IMV isolation, while the level of contaminants should decrease with isolation. The progressive diminution of Na⁺/K⁺-ATPase activity and WB signal with the isolation steps is similar to that reported by Norton and Poduslo (1973). Finally, the WB signal of MBP, representing 30% of total proteins in myelin sheath, increased with the grade of IMV isolation, while the presence of axolemma seems limited (Fig. 1 and Table 1).

Two new acquisitions follow, which sound interesting in the general biology field: (i) myelin may represent a site of consistent aerobic metabolism, probably to supply the energetic demand of axon; (ii) mitochondria would not be the exclusive site of aerobic ATP synthesis. Several studies have identified axonal degeneration as the major determinant of irreversible neurological disability in many demyelinating diseases, but the mechanisms involved are poorly understood. Among the proposed hypotheses for degeneration of demyelinated axons (inflammatory causes; accumulation of Ca²⁺) a lack of trophic support by myelin and an energy depletion was proposed (Correale et al., 2006). Supposing that the trophic role exerted by myelin is to carry out an aerobic ATP synthesis, demyelination would cause ATP depletion, which in turn would lead to axonal degeneration.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.biocel.2009.01.009.

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